

Crip Making

AIMI HAMRAIE

During the COVID-19 pandemic, many people turned to remote and digital forms of participation and communication, whether for work or social life. What many nondisabled people did not know was that disability communities had often developed (for themselves) the same technologies that many nondisabled people were using to survive the pandemic—despite often being denied the right to remote work and education. At the same time, Black, Indigenous, Latinx, and institutionalized disabled people have been at particularly high risk for severe COVID-19 complications, including death, and were deprioritized in medical triage and vaccination. Disability communities responded to these paradoxes of access in several ways. Some launched hashtag campaigns. For example, Alice Wong’s #HighRiskCA pointed out that disabled people often had less access to vaccinations, while Kate McWilliams’s #AccessibilityforAbleDs offered examples of how remote access for nondisabled people was being prioritized during the pandemic. Others hosted online cultural events with ASL interpreters, live transcription, and image descriptions. Still others created digital networks for mutual aid and the distribution of items such as masks, hand sanitizer, and air filters. In the process, disability communities claimed expertise and design knowledge about remote forms of participation. Their practices illustrate the concept of “disability culture,” comprising the norms, social relations, and technologies that form around collective experiences of disability (Barnes and Mercer 2001). Whether in the case of “homebound shut-ins” embracing radio technology in the 1920s (Kirkpatrick 2017), polio survivors sharing design hacks via newsletters in the 1960s (Williamson 2012), Autistic people finding community via email listservs in the 1990s (Sinclair 2012), chronically ill people organizing funerals and protests from their beds (Piepzna-Samarasinha 2018), or Black disabled feminists using “hashtag activism” for political advocacy (Thompson 2019; S. Jackson, Bailey, and Welles 2020), remote access has been central to the socio-technological life of disability in the twentieth and twenty-first centuries, highlighting the reaches of geographically dispersed and heterogeneous disability cultures.

Yet technology is not an easy fix. Disabled critics of technology highlight the role of “technoableism” (Shew 2020) in shaping the treatment of disability as a deficiency or problem to fix. Crip theorists note disabled peoples’ “ambivalent relationships to technology” (Kafer 2013, 119), in which technological failures

and conflicting accessibility needs create frictions. For example, remote learning that benefits some chronically ill people may cause “Zoom fatigue” or chronic migraines for the same people or for others. Telehealth and telework options during the pandemic are being rescinded in light of mass vaccinations, leaving chronically ill people without accommodations and sparking protests (Kovach 2011). Useful digital tools and electronics (including internet routers and ventilators) sometimes fail in locations facing power shortages due to heat, wildfires, or inadequate infrastructure. The many paradoxes of remote access illustrate that disabled people are makers and adapters, and not just passive users, of technology. Rather than eschewing technology, disability communities often turn to iterative design processes to address these apparent conflicts.

The politics of disability technology raise important theoretical and empirical questions about the epistemologies, methodologies, and societal implications of design *by*, rather than *for*, disabled people. In many cases, *making* is political in that it gathers networks of kinship and belonging around world-building practices that challenge mainstream cultural norms and values. Critical making theorist Jentery Sayers (2017) argues that making is a way of drawing boundaries: “Such boundaries mark how and where value is attributed and accrued. They also correspond with ideologies and environments for who gets to make, who can maintain, and who must source” (9). Consequently, *crip making* relies on and produces forms of knowledge and expertise that draw boundaries between assimilation and antiassimilation, able-bodied norms and disability culture. In this chapter, I explore the concept of crip making in three ways. First, I examine controversies surrounding the role of the maker (as author or progenitor) in debates about disability, design, and technology. Then, I examine political-relational and performance-based approaches to disability-led design, looking at the design politics of lived experience and complicating notions of “standpoint epistemology” and “situated knowledge.” Finally, I draw on an example of crip making during the COVID-19 pandemic to highlight the complex politics of access friction and negotiation.

an issue that
shivers-McNair
likewise
exposed

Users and Makers: Disability, Design, and Technology

Disability has been a present, but often unrecognized, force in user-centered design. Most often, disability has been treated as a problem to avoid or solve through technology. Nineteenth- and twentieth-century designers treated impairment as a confounding element in industrial systems and prescribed technological solutions focused on normalization and assimilation (Hamraie 2017; Serlin 2004). For example, artificial limbs combined functional and aesthetic considerations to make users appear typical. But disability also shapes contemporary and mainstream technologies (including remote-access technologies) in ways that are often latent or undocumented (Hendren 2020; Williamson

2012; Williamson and Guffey 2020). For example, Deaf people were historically viewed as testing sites for telephonic technologies and electronics that were then adapted into the mainstream (Mills 2011). Archivist and material culture historian Katherine Ott (2014) refers to this latency, and the broader circulation of disability-generated technologies, as “disability things,” a concept that describes the material culture of disability as expanding beyond the body into built and social worlds.

Scholars document technological biases against disabled people. Critical scholarship has formed a consensus that technology (whether assistive, physical, or digital) is laden with assumptions about disability as an undesirable condition in need of normalization (Alper et al. 2015; Moser 2006; Gibson 2006). These assumptions undergird a type of inequality that disabled philosopher of technology Ashley Shew (2020) terms “technoableism.” Corporate disability “hack-a-thons,” for example, frequently treat disability as a problem in need of solving and enlist disabled people as user-experts, not designers (Wong 2015; Yergeau 2014). Critics reject “technological saviorism,” or the assumption that corporations and engineers—often nondisabled people—can liberate or enable disabled people through technology. For example, public historian and designer Liz Jackson (2019) describes “disability dongles” as well-intentioned technologies, such as stair-climbing wheelchairs or ASL gloves, that do not meet disabled users’ actual needs or desires. In response, sociologists and philosophers of technology document disabled lived experiences as distinct from engineers’ perceptions of disability (Shew 2020).

Disability studies scholars point out that technoableism and saviorism have not deterred disabled people from using technology, however. “Dismediation” theories of disability and digital media simultaneously engage with and trouble lived experiences of disability in order to “resist rehabilitation and standardization” (Mills and Sterne 2017, 365). Scholars propose instead that disabled people have “ambivalent relationships to technology” (Kafer 2011, 119), wherein users may adopt imperfect, uncomfortable, or less functional tools, even while critiquing them. This observation aligns with Lewis Mumford’s classic assertion in the field of the philosophy of technology of the “ambivalence of the machine” (1934, 283), which simultaneously liberates and harms. Feminist science and technology studies (STS) scholars have likewise framed technology as “non-innocent,” simultaneously produced by systems of oppression and in opposition to them (Haraway 1991). An understanding of technological ambivalence and non-innocence leads disability historians to point out that using imperfect technologies emboldens disabled users as design experts; likewise, disabled people have contributed to designing assistive technologies, in addition to using them (Virdi 2020; Ott, Serlin, and Mihm 2002; Williamson 2012; Hendren and Lynch, n.d.).

Disabled-led design is thus both an individual practice and a collective phenomenon illustrating disability culture. In contrast to technoableism, Alice

though focused on the context of political repression and surveillance in pedagogical contexts, Willard (2025) offers a really fruitful account of ambivalence for theorizing this “technological ambivalence.”

Sheppard (2019) argues that disability technology can emerge through a “body of knowledge . . . and a way of producing, developing, sharing, and accessing it . . . that are held both individually and in community.” This communal, collective, and cultural understanding of disability rests on notions of aesthetic and relational community as emerging from disability culture.

Crip Making: Political-Relational, Disabled-Led Design

In their 1979 book, *Design for Independent Living*, nondisabled architects Raymond Lifchez and Barbara Winslow translate for other nondisabled architects the philosophies of the radical disability movement in Berkeley, California. Based on their close work with this movement in a design studio taught at UC Berkeley’s College of Environmental Design, and further collaborations with the Center for Independent Living, Lifchez and Winslow transmit a sentiment that to most architects, rehabilitation professionals, and other nondisabled people would appear to be a bold assertion: that disabled people are not trying to be like everyone else, and in fact, they are changing the world by making things that refuse assimilation and conformity. In other words, disabled people are making things to make it in an inaccessible world.

In the book’s epilogue, Lifchez and Winslow ask,

Is the objective to assimilate the disabled person into the environment, or is it to accommodate the environment to the person? . . . Currently, the emphasis is on assimilation, for this seems to assure that the disabled person, once “broken-in,” will be able to operate in a society as a “regular person” and that the environment will not undermine his natural agenda to “improve” himself. As we have shown, this assumption can be counter-productive when designing for accessibility. It may serve only to obscure the fact that the disabled person may have a point of view about the design that challenges what the designers would consider good design. (1979, 150)

making is a rhetorical resource employed by disabled people, much like the rhetorical lowering of artistic crip time that Saunders (2020) describes

The book documents this antiassimilationist worldview through photographs, design documentation, and narratives of the often-quotidian individual and collective tools that disabled people create. In doing so, it offers a snapshot of disability cultural approaches to technology. For example, in spaces shared by disabled people, where disabled people are not required to appear “normal,” Lifchez and Winslow (1979, 51) write that there is “shared understanding” of access barriers and even “crip humor” regarding bodily differences and experiences. This early use of “crip” in relation to making and designing reveals the parallel and intersecting trajectories of antinormative disability philosophies and practices of making things (and thus “drawing boundaries,” to return to Sayers) that disability introduces.

pointing to the need for pluriversal understandings of design (Escobar, 2018)

Movements of physically disabled people emerged in the 1960s and 1970s to challenge the primacy of physically nondisabled people in shaping the norms to which disabled people are held. Since that time, the word *crip* has expanded in its meanings, taken up as a philosophy of disability that agitates against compulsory normalcy of any kind, whether able-bodiedness or able-mindedness. Some have “claimed crip” in a similar manner to the term *feminist* (Schalk 2013). Many of these debates about the ontology of crip also address its epistemic status—who is an expert, who gets to claim to know—as determining its material and ethical stakes. Crip “onto-epistemology” (Barad 2007) can thus be understood as bound up in political questions of who gets to “know” disability and design in the name of disabled people. As in my broader work (Hamraie 2013; Hamraie 2017), I commit here to an additional step: that crip being and knowing are inseparable from crip ways of making and acting. In other words, crip “knowing-making” (Hamraie 2017) is a commitment to right action through changing existing material arrangements, not endorsing the nondisabled, assimilationist norm.

Since the early 2000s, crip theorists have addressed cultural, philosophical, and political dimensions of antiassimilation (Sandahl 2003; McRuer 2006; Kafer 2011). Crip making is implied within (but not often explicitly named in) these theories. That is, *crip* is not a synonym for *disability*, nor is it simply a political orientation. Rather, it is a specific commitment to shifting material arrangements, or what Robert McRuer names “the will to remake the world” (McRuer 2006, 35). Bess Williamson and Elizabeth Guffey’s “design theory of disability” (2020) likewise stipulates that disability is a generative, world-changing phenomenon, not a deficit. Because most research on disability and technology presumes that disabled people are users, rather than makers and designers, new frameworks (such as crip technoscience) prioritize disabled-led design and making (Hamraie 2017; Hamraie and Fritsch 2019). Crip technoscience manifests in technological hacks, adaptations, or innovations emerging from what Leah Lakshmi Piepzna-Samarasinha (2018) terms “crip science,” using the disability community’s reclamation of the word *cripple* to define a locus of valuable expertise about navigating and adapting to inaccessible worlds. As the term *crip* has been reclaimed, it has grown into a field of scholarship (crip theory) and describes commitments to interdependence, collectivity, and anticapitalism, similar to the disability justice framework (Kafer 2011).

The emerging field of crip technoscience studies (Fritsch et al. 2019) addresses dimensions of disability technology that go beyond mechanical functionality or aesthetic assimilation to examine the cultural politics of technology, including how claims of expertise operate as forms of power that shape both knowing and designing. Thus, crip technoscience is not a “standpoint epistemology” (Harding 1992), a feminist and Marxist concept that positions the perspective of the oppressed as uniquely true or objective. Nor is crip technoscience a practice that derives solely from “disability expertise” (Hartblay 2020) or authorship. Not

reminiscent
again of Escobar's
(2016) ontological
design and investment
in rejecting a
singularity prescribed
highly
onto-epistemologic
order

which has some
resonance with
understanding
decolonial as not
a mere metaphor,
but a material +
active move toward
Indigenous sovereignty,
reparation, + Indigenous
world-building
Tuck + Yang, 2012;
Khuon, 2021; Mathison, 2021

everything made by a disabled person is crip technoscience. Beyond knowledge itself, crip technoscience provides a framework for studying disability design with specific political commitments, specifically resistance to imperatives for normalization and assimilation (Hamraie and Fritsch 2019; Gotkin 2019; Sheppard 2019; Nelson et al. 2019; Wong 2019; Thompson 2019).

Crip technoscience theory speculates that disabled forms of creative ingenuity, resourcefulness, hacking, and trial and error are not only scientific or epistemic but *technoscientific*—a term that STS scholars use to describe the mutual reliance of knowledge production and technological innovation (Hottot 2018; Latour 2005; Haraway 1997). Justice-centered feminist and decolonial technoscience studies have taken up these latter meanings to examine the roles of race, gender, and the nation-state in framing technology (Murphy 2012; Subramaniam et al. 2016) and have accordingly offered both critiques and alternatives rooted in political frameworks devoted to unsettling science. A crip technoscience theoretical framework builds on feminist and decolonial technosciences to differentiate between technologies developed through models of disability-as-pathology (often in the name of rehabilitating injured soldiers) and those derived from disability culture communities, where technology supports embodied differences and interdependent socialities.

STS scholars in “user studies” have addressed the figure of the user and how it both configures and is configured by technology (Oudshoorn and Pinch 2003). Configuration names ontologies and epistemologies of use, or how design shapes the parameters of embodied interaction with technologies. Disability scholars and activists claim expert use as a form of design knowledge. For example, disability communities discuss disability-led design with frequency, with some declaring, “We are the original lifehackers” (L. Jackson 2018). In this context, studies of disability technology argue that disabled people are user-experts, who engage in “microactivist affordances” (Dokumaci 2016) and “disability hacktivism” (Yergeau 2014).

Since disability movements in the 1970s and 1980s made changes to the built environment their focus, philosophers of technology have shown interest in disabled peoples’ critiques of inaccessible design as examples of the democratic politics of technology. Some philosophers have treated disability activists as prototypical user-experts, whose political advocacy stemming from lived experiences of disability shapes material arrangements (Winner 1980; Feenberg 2012). In all of these treatments of disabled people as makers, do-it-yourself capacities are largely taken for granted as radical and democratic. However, questions remain about *how* the lived experience of disability, as well as participation in disability culture, shapes design processes. What forms of knowledge matter, and how do they result in design decisions or material changes?

Crip making draws on STS approaches to expert knowledge and material practice, or “knowing-making” (Hamraie 2017), to inquire about how remote-access

methodologies can inform understandings of community-generated “collective access” (Mingus 2010). It builds on disability studies approaches that treat disability as a “political-relational” (Kafer 2011) and interdependent (Berne et al. 2018) (rather than individual) phenomenon through which technologically mediated cultural collaborations yield new material arrangements. Drawing on elements of these literatures while following Black disabled scholar and wheelchair dancer Alice Sheppard’s call to study the “cultural-aesthetic” dimensions of disability technology (Sheppard 2019), crip making examines the role of disability *community expertise* in shaping remote access.

Crip making highlights the central role of design frictions in shaping disabled relationships to technology. *Design friction* names the practices of tinkering and hacking that provide imperfect access or highlight conflicting access needs while simultaneously building sociality and interdependence (Hamraie and Fritsch 2019). For example, wheelchair users and blind people have historically had different needs regarding sidewalk construction because the same curb ramps that allow wheelchair users to move freely from sidewalk to street create dangerous situations for blind people who do not know a street is coming. However, activists in both communities approached conflicting access needs to develop “tactile paving,” the raised yellow bumps that appear on sidewalk corners to announce the elevation change (Hamraie 2017). A material instance of friction thus came to exemplify the forms of conflict and negotiation that frame disability design.

Crip making also relates to the emerging frameworks of disability justice and design justice. Disability justice is a political framework that highlights the “leadership of the most impacted,” “interdependence,” and “collective access” (Berne et al. 2018). As discussed earlier, this movement is led by disabled people of color and collectives drawing on art and design to do political advocacy. Disability justice thus has a significant focus on the material manifestations of political values, in addition to creating a new theoretical framework for disability politics. However, there is not yet adequate empirical documentation of disabled designers of color and their contributions to shaping conceptions of disability and design justice. Similarly, *design justice* (Costanza-Chock 2020) is an emerging framework from within design discourse that challenges the charitable sensibilities of technology designed for users, and instead devises methods for prioritizing marginalized users. Crip making synthesizes the two frameworks into “disability design justice” (Hamraie 2020), naming and building on an existing field of protocols and theories.

Remote Access: Crip Making in Pandemic Times

If disability-led design does not seek quick fixes or technological saviorism, how does this phenomenon instead illustrate community knowledge and collaboration? To answer this question, we need to understand the meanings of *access* as related to making and design.

Access is often understood as a spatial phenomenon. Twentieth-century efforts to deinstitutionalize disabled people often focused on creating accessible public spaces in order to support integration into the community. As a result, the concept of accessibility is often contrasted with the isolation of institutionalization; consequently, disability scholars theorize “publicity as capacity,” a type of power derived from being able to be present (Johnson 2020). Likewise, critical scholarship on accessibility has typically focused on technologies that enable physical presence in built environments (Guffey 2017; Hamraie 2017; Titchkosky 2011; Williamson 2019). Furthermore, rich ethnographic scholarship on digital accessibility practices and negotiations (Ellcessor 2006) and “digital activism” (e.g., McLean 2020; Thompson 2019) has interrogated notions of participation and use, but addresses digital spaces rather than technologies and design practices.

a timely issue, given the recent DOJ memo under the Trump administration suggesting that the DMCA is L.C. (1999) ruling does not require states to provide services as alternatives to institutionalization

Yet many disabled people have required physical distancing for safety, accessibility, and well-being. This is particularly true for chronically ill, chemically sensitive, asthmatic, Deaf and hard of hearing, blind, and neurodivergent people. The widespread use of masks, social distancing, text-based internet communication, and video conferencing during the pandemic follows on practices that disabled communities have designed. Remote access, in turn, raises questions about how distance, rather than proximity, shapes disability culture. Remote access thus highlights how technological mediation creates or forecloses “cultural locations of disability” (Snyder and Mitchell 2010) through dispersed geographies of technoscience (Livingstone 2010). A cripping framework inquires about how physical distance—such as living in different cities or away from community, or being unable to share space without masking due to COVID-19—can also be an enabling condition for technological design and participation, while continually requiring adaptations toward more workable solutions.

Remote access is inherently frictioned and noninnocent in that the same conditions enabling access for some can create hierarchies of belonging for others. Distance can be a disabling condition with important implications for disability world-making. For example, John Lee Clark offers a deafblind critique of “distantism,” or “privileging of the distance senses of hearing and vision,” which, unlike tactile senses, tend to operate across wider spaces (Clark 2017). Clark extends the critique of distantism to paternalistic attitudes and behaviors, such as assistants who presume to be experts about what deafblind people need. In one story, Clark recounts an assistant who accompanies him to the grocery store:

She’s confused when I do not give her a list and I take charge, directing us toward the places where we will find what I want. She is now more like a detector, or a device that I take out of my back pocket to consult. Only she is far more intelligent than any machine could be and there’s a wonderful rapport—that is, if she is able to unlearn her distantism. It is my responsibility to learn and know the world around

me; it is part of her job to help me update that knowledge as we go along. But it is not her job to retain any of this knowledge herself. (Clark 2017)

In this example, Clark responds to distantist paternalism by asserting expertise and building communal rapport. Though not an explicit act of *making*, Clark's response configures the assistant as a kind of "device" who helps with shopping but within whom expertise is not expected to reside.

Clark's pivot and reconceptualization recall other practices of participatory ³ access-making. Unlike assistive technologies focused on individual users, remote access frequently emphasizes mass technologies, in which communities participate in coproducing material norms and practices. Listservs and discussion groups emerged on the internet that were devoted to specific disability communities, such as chronically ill "spoonies" and neurodivergent people. In the 2010s, remote access became a norm within "cross-disability" spaces, which included people with more than one type of disability. Crip making illustrates remote access in disability culture, including activist meetings conducted from beds with phones during the Occupy at Home movement, live webstream performances with ASL and image description, Zoom parties, and text-only email courses and Twitter conferences. Emerging archival projects document these practices, building on earlier research regarding histories of disability and print media, such as letters, newsletters, telegrams, and phone trees (e.g., Burch 2021; Williamson 2012).

Digital and internet-enabled forms of accessibility have shifted the availability of access labor: whereas previously, ASL interpreters and captioners had to be available in person, new technologies such as video conferencing enable streaming access services. Artificial intelligence transcription services such as Otter.ai enable broader accessibility, but also displace paid labor for access providers and often display racial and gender bias (Hickman 2019). "Mobile technologies," such as the iPad, have promised access to language through assisted communication, but also produced inequalities of race, gender, and class (Alper 2017). Likewise, online digital communities enable disabled people to connect across long distances, but create inequalities within and among disability communities, such as when videos do not have closed captioning or websites are not built for screen readers, a technology used by blind people (Ellcessor 2016). Disability communities respond by exploring new design solutions. These points highlight considerations of how disability culture forms through "design friction" (Forlano and Mathew 2014), rather than consensus or heterogeneity.

Examples of
non-innocence

The feminist STS concept of "protocol" (Murphy 2012) describes methodological practices that become both standardized and reiterated in pursuit of particular political goals. Crip making adopts protocol, alongside expert knowledge, as a site of inquiry into design methodologies more generally. Because disability communities comprise diverse individuals with different access needs, the

design protocols for remote access are constantly being developed and rewritten. Emergent practices include developing protocols for rich textual descriptions of visual content (Finnegan and Coklyat, n.d.); hacking Zoom to integrate live captioning and audio description (Gotkin, Hickman, and Hamraie 2020); and developing apps that afford unique sensory, artistic, and even poetic experiences (Sheppard 2019). While these protocols purport to embody the disability justice principle of interdependence, questions remain about how lived and communal disability knowledge shaped them, and the technologies for which they create access. Remote access illustrates how such practices emerged, often in response to physical distance and technological mediation.

Protocols for Remote Access: Experiments in Crip Making

On a cold March afternoon in 2020, I turned on my laptop and clicked a URL to join a Zoom call. With my phone, I dialed into a different Zoom call, where I would be providing live audio description for the next hour. My computer screen revealed a grid of twenty faces, people seated before a screen. In the first image, a DJ sat at their table with mixing equipment and a speaker, backlit with pink light. In other boxes, partygoers wore outfits covered in glitter and sequins and swayed their bodies to slow tunes. A chatbox next to the images offered descriptions of the mood and speed of the music. Lyrics and spoken language were transcribed in on-screen captioning. An ASL interpreter signed the lyrics and other words. A team of volunteers worked as “access doulas” in the chat section, sparking conversation, describing sound, naming the songs, and making fun party banter.

The event was Remote Access: A Crip Nightlife Party, hosted by Critical Design Lab, a collaborative of disabled designers, artists, and researchers. Disability scholar and organizer (and lab member) Kevin Gotkin (aka DJ Who Girl) had been writing about histories of disability nightlife (Gotkin 2019). We started organizing the party as the COVID-19 pandemic took our teaching, organizing, and socializing online. We wanted to show what disabled communities were already doing with remote and digital forms of participation, as well as to create a space for our communities to share movement, celebration, and kinship in the midst of grief.

The party sought to bring elements of crip digital culture into a space of celebration and aesthetic exploration. Alongside the DJ set, Indigenous disabled artist moira williams's camera focused on natural materials—soil, rocks—in front of their screen. Taiwanese American disabled artist Yo-Yo Lin screen-shared work studying body movements and joints popping—a computerized image of a body moving in space, surrounded by orbs of neon light. The material, the elemental, the earthly, the embodied, and the digital married in an ecstatic jubilee of togetherness and belonging as DJ Who Girl mixed together popular dance music with slower vibes.

Muting my computer, I took my role as the live image describer by speaking into my phone. For the next hour, I described visual content while I danced along to the music and waved at my friends on the screen. When my shift was over, another participant took over. Many other participants did the same work via text in the chat box. On their own screens, participants danced and played. Sky Cubacub—a Filipinx queercrip fashion designer known for their neon spandex designs—danced in their studio with their mother. Cubacub wore a self-designed chainmail cap and a silver hologram outfit reminiscent of David Bowie, while their mother wore a long, flowing, neon skirt. In other screens, people danced with their hands, some voguing. Some wore costumes or masks, dressed as cats, or showed up in neon-pink furs. Others just sat and watched or listened to the music. Some showed their pets and other companions. Some sat outdoors.

Remote Access was not merely a celebratory event, however. It was also part of a series of research and design projects centered in the Critical Design Lab. By design, and due to access needs, physical distance, and living across many time zones, the lab already conducts all of our work remotely: we meet and host workshops using digital conferencing platforms with integrated captioning, and we create media (such as a podcast) and curate exhibitions using digital tools. But a Zoom party held for potentially hundreds of disabled people and allies also presented new challenges. The organizing team met to discuss what it would take to make an online party accessible. The graphics needed to be image-described and high contrast, legible for people with color-blindness. Though they were abstract and dynamic, the art visuals needed to have clear descriptions. The Zoom platform itself also had to be tweaked. Because Zoom only picks up on sound in the human voice range, the music did not sound very bass-heavy through the computer speakers. Also Zoom’s chat box, if relied on for image descriptions, would mute out the sound for screen-reader users, meaning that blind people would not be able to hear the descriptions of the screen at the same time as the music.

The party thus became a site of design experimentation through crip making. We began by designing participation in the event with a protocol (Gotkin, Hickman, and Hamraie 2020), sharing it with many collaborators, incorporating feedback, and further designing the many ways of accessing information and aesthetics. In other words, the party itself (and the party-planning process) served as a design charette, an opportunity for real-time creative experimentation with the meanings of access. The “access doula” team (named by Gotkin), assembled from disabled artists and scholars we knew, formed a crucial piece of the technological puzzle. Working in shifts, we engaged with technology as a translational apparatus for also generating new poetics of crip culture and community. These descriptions and translations also appeared alongside party chatter, a digitally enabled form of sociality fostering collective belonging in the space.

Despite the pervasiveness of both social isolation and remote access in disability culture, many of our cultural spaces have been affected by the COVID-19

pandemic. Remote Access has been an ongoing party series, with the intention of further reiterating and making anew the social worlds that disability generates. But COVID-19 is also mobilizing disabled people to do what we have been doing: organizing mutual aid under difficult circumstances. **Mutual aid** is a central tenet of the disability justice movement, led by disabled people of color and queer disabled people, and enacted through principles of collective access and liberation, anticapitalist politics, and leadership of the most affected (Berne et al. 2018). As the pandemic continues, mutual aid can look like saving lives and reducing harm. But it can also take shape as creating crip joy, holding space for crip pain, and forging networks of solidarity to break isolation. Remote Access is thus more than a dance party. It is an enactment of crip making through disability culture as a political act of mutual aid.

BIBLIOGRAPHY

- Alper, Meryl. 2017. *Giving Voice: Mobile Communication, Disability, and Inequality*. Cambridge, MA: MIT Press.
- Alper, Meryl, Elizabeth Ellcessor, Katie Ellis, and Gerard Goggin. 2015. "Reimagining the Good Life with Disability: Communication, New Technology, and Humane Connections." In *Communication and the "Good Life,"* edited by Hua Wang, 197–212. New York: Peter Lang.
- Barad, Karen. 2007. *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning*. Durham, NC: Duke University Press, 2007.
- Barnes, Colin, and Geoff Mercer. 2001. "Disability Culture: Assimilation or Inclusion." In *Handbook of Disability Studies*, edited by Gary L. Albrecht, Katherine D. Seelman, and Michael Bury, 515–534. Thousand Oaks, CA: Sage.
- Berne, Patricia, Aurora Levins Morales, David Langstaff, and Sins Invalid. 2018. "Ten Principles of Disability Justice." *WSQ: Women's Studies Quarterly* 46 (1): 227–230.
- Burch, Susan. 2021. *Committed: Remembering Native Kinship in and beyond Institutions*. Chapel Hill: University of North Carolina Press.
- Clark, John Lee. 2017. "Distantism." *Wordgathering* 11 (3). https://wordgathering.com/past_issues/issue43/essays/clark.html.
- Costanza-Chock, Sasha. 2020. *Design Justice: Community-Led Practices to Build the Worlds We Need*. Cambridge, MA: MIT Press.
- Dokumaci, Arseli. 2016. "Micro-activist Affordances of Disability: Transformative Potential of Participation." In *ReClaiming Participation*, edited by Mathias Denecke, Anne Ganzert, Isabell Otto, and Robert Stock, 67–84. Bielefeld, Germany: Transcript Verlag.
- Ellcessor, Elizabeth. 2016. *Restricted Access: Media, Disability, and the Politics of Participation*. New York: New York University Press.
- Feenberg, Andrew. 2012. *Questioning Technology*. London: Routledge.
- Finnegan, Shannon, and Bojanna Coklyat. n.d. Alt-Text as Poetry homepage. Accessed October 11, 2022. <https://alt-text-as-poetry.net/>.
- Forlano, Laura, and Anijo Mathew. 2014. "From Design Fiction to Design Friction: Speculative and Participatory Design of Values-Embedded Urban Technology." *Journal of Urban Technology* 21 (4): 7–24.
- Fritsch, Kelly, Aimi Hamraie, Mara Mills, and David Serlin. 2019. "Introduction to Special Section on Crip Technoscience." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/31998>.

- Gibson, Barbara E. 2006. "Disability, Connectivity and Transgressing the Autonomous Body." *Journal of Medical Humanities* 27 (3): 187–196.
- Gotkin, Kevin. 2019. "Crip Club Vibes." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/30477>.
- Gotkin, Kevin, Louise Hickman, and Aimi Hamraie. 2020. "Remote Access: Crip Nightlife Participation Guide." With the Critical Design Lab. March 2020. <https://docs.google.com/document/d/1vluVl73ZZLgMNX6qvXgjbZnlO7yYoQUioJpvJ8Jh-Q/edit>.
- Guffey, Elizabeth. 2017. *Designing Disability: Symbols, Space, and Society*. London: Bloomsbury.
- Hamraie, Aimi. 2013. "Designing Collective Access: A Feminist Disability Theory of Universal Design." *Disability Studies Quarterly* 33 (4). <https://dsq-sds.org/article/view/3871/3411>.
- . 2017. *Building Access: Universal Design and the Politics of Disability*. Minneapolis: University of Minnesota Press.
- . 2020. "From Exceptional Accommodations to Disability Justice Design: Ways of Thinking about Accessibility as Hospitality." In *Rehearsing Hospitalities Companion 2*, edited by Yvonne Billimore and Jussi Koitela, 113–126. Helsinki: Ministry of Education and Culture, Finland, and Archive Books.
- Hamraie, Aimi, and Kelly Fritsch. 2019. "Crip Technoscience Manifesto." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/29607>.
- Haraway, Donna. 1987. "A Manifesto for Cyborgs: Science, Technology, and Socialist Feminism in the 1980s." *Australian Feminist Studies* 2 (4): 1–42.
- Haraway, Donna. 1991. *Simians, Cyborgs, and Women: The Reinvention of Nature*. New York: Routledge.
- Harding, Sandra. 1992. "Rethinking Standpoint Epistemology: What Is 'Strong Objectivity?'" *Centennial Review* 36 (3): 437–470.
- Hartblay, Cassandra. 2020. "Disability Expertise: Claiming Disability Anthropology." *Current Anthropology* 61 (S21): S26–S36.
- Hendren, Sara. 2020. *What Can a Body Do? How We Meet the Built World*. New York: Penguin.
- Hendren, Sara, and Catrin Lynch. n.d. Engineering at Home homepage. Accessed October 11, 2022. <http://engineeringathome.org/>.
- Hickman, Louise. 2019. "Transcription Work and the Practices of Crip Technoscience." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/32081>.
- Hottois, Gilbert. 2018. "Technoscience: From the Origin of the Word to Its Current Uses." In *French Philosophy of Technology: Classical Readings and Contemporary Approaches*, edited by Sacha Loeve, Xavier Guchet, and Bernadette Bensaude-Vincent, 121–138. Cham, Switzerland: Springer.
- Jackson, Liz. 2018. "We Are the Original Lifehackers." *New York Times*, May 30, 2018.
- . 2019. "Disabled People Want Disability Design—Not Disability Dongles." CBC Radio, November 8, 2019.
- Jackson, Sarah J., Moya Bailey, and Brooke Foucault Welles. 2020. *#HashtagActivism: Networks of Race and Gender Justice*. Cambridge, MA: MIT Press.
- Johnson, Jenell. 2020. "Breaking Down: On Publicity as Capacity." *Rhetoric Society Quarterly* 50 (3): 175–183.
- Kafer, Alison. 2013. *Feminist, Queer, Crip*. Bloomington: Indiana University Press.
- Kirkpatrick, Bill. 2017. "A Blessed Boon." In *Disability Media Studies*, edited by Elizabeth Ellcessor and Bill Kirkpatrick, 330–356. New York: New York University Press.
- Kovach, Sydney. 2011. "Disabled Student Union Hosts Hybrid Protest for COVID-19 Accommodations." *Daily Bruin*, October 11, 2011. <https://dailybruin.com/2021/10/09/disabled-student-union-hosts-hybrid-protest-for-covid-19-accommodations>.

- Latour, Bruno. 2005. *Reassembling the Social: An Introduction to Actor-Network-Theory*. Oxford: Oxford University Press.
- Lifchez, Raymond, and Barbara Winslow. 1979. *Design for Independent Living: The Environment and Physically Disabled People*. Berkeley: University of California Press.
- Livingstone, David N. 2010. *Putting Science in Its Place*. Chicago: University of Chicago Press.
- McLean, Jessica. 2020. "'It's Just Coding': Disability Activism In, and About, Digital Spaces." In *Changing Digital Geographies: Technologies, Environments and People*, 229–246. Cham, Switzerland: Palgrave Macmillan.
- McRuer, Robert. 2006. *Crip Theory: Cultural Signs of Queerness and Disability*. New York: New York University Press.
- Mills, Mara. 2011. "Deafening: Noise and the Engineering of Communication in the Telephone System." *Grey Room* 43:118–143.
- Mills, Mara, and Jonathan Sterne. 2017. "Afterword II: Dismediation: Three Propositions and Six Tactics." In *Disability Media Studies: Media, Popular Culture, and the Meanings of Disability*, edited by Elizabeth Ellcessor and Bill Kirkpatrick, 365–378. New York: New York University Press.
- Mingus, Mia. 2010. "Reflections from Detroit: Reflections on an Opening: Disability Justice and Creating Collective Access in Detroit." INCITE blog, August 23, 2010. <https://incite-national.org/2010/08/23/reflections-from-detroit-reflections-on-an-opening-disability-justice-and-creating-collective-access-in-detroit/>.
- Moser, Ingunn. 2006. "Disability and the Promises of Technology: Technology, Subjectivity and Embodiment within an Order of the Normal." *Information, Communication and Society* 9 (3): 373–395.
- Mumford, Lewis. 1934. *Technics and Civilization*. Chicago: University of Chicago Press.
- Murphy, Michelle. 2012. *Seizing the Means of Reproduction*. Durham, NC: Duke University Press.
- Nelson, Mallory Kay, Ashley Shew, and Bethany Stevens. 2019. "Transmobility: Possibilities in Cyborg (Cripborg) Bodies." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/29617>.
- Ott, Katherine. 2014. "Disability Things: Material Culture and American Disability History, 1700–2010." In *Disability Histories*, edited by Susan Burch and Michael Rembis, 119–135. Urbana: University of Illinois Press.
- Ott, Katherine, David Serlin, and Stephen Mihm, eds. 2002. *Artificial Parts, Practical Lives: Modern Histories of Prosthetics*. New York: New York University Press.
- Oudshoorn, Nelly, and Trevor Pinch, eds. 2003. *How Users Matter: The Co-construction of Users and Technology*. Cambridge, MA: MIT Press.
- Piepzna-Samarasinha, Leah Lakshmi. 2018. *Care Work: Dreaming Disability Justice*. Vancouver, Canada: Arsenal Pulp.
- Sandahl, Carrie. 2003. "Queering the Crip or Crippling the Queer? Intersections of Queer and Crip Identities in Solo Autobiographical Performance." *GLQ: A Journal of Lesbian and Gay Studies* 9 (1): 25–56.
- Sayers, Jentery. 2017. "Introduction: I Don't Know All the Circuitry." In *Making Things and Drawing Boundaries: Experiments in the Digital Humanities*, edited by Jentery Sayers, 1–17. Minneapolis: University of Minnesota Press.
- Schalk, Sami. 2013. "Coming to Claim Crip: Disidentification with/in Disability Studies." *Disability Studies Quarterly* 33 (2). <https://dsq-sds.org/article/view/3705/3240>.
- Serlin, David. 2004. *Replaceable You: Engineering the Body in Postwar America*. Chicago: University of Chicago Press.
- Sheppard, Alice. 2019. "Staging Bodies, Performing Ramps." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/30459>.

- Shew, Ashley. 2020. "Ableism, Technoableism, and Future AI." *IEEE Technology and Society Magazine* 39 (1): 40–85.
- Sinclair, Jim. 2012. "Don't Mourn for Us." *Autonomy, the Critical Journal of Interdisciplinary Autism Studies* 1 (1).
- Snyder, Sharon L., and David T. Mitchell. 2010. *Cultural Locations of Disability*. Chicago: University of Chicago Press.
- Subramaniam, Banu, Laura Foster, Sandra Harding, Deboleena Roy, and Kim TallBear. 2016. "Feminism, Postcolonialism, Technoscience." In *The Handbook of Science and Technology Studies*, 4th ed., edited by Ulrike Felt, Rayvon Fouché, Clark A. Miller, and Laurel Smith-Doerr, 407–434. Cambridge, MA: MIT Press.
- Thompson, Vilissa. 2019. "How Technology Is Forcing the Disability Rights Movement into the 21st Century." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/30420>.
- Titchkosky, Tanya. 2011. *The Question of Access: Disability, Space, Meaning*. Toronto: University of Toronto Press.
- Virdi, Jaipreet. 2020. *Hearing Happiness: Deafness Cures in History*. Chicago: University of Chicago Press.
- Williamson, Bess. 2012. "Electric Moms and Quad Drivers: People with Disabilities Buying, Making, and Using Technology in Postwar America." *American Studies* 52 (1): 5–29.
- . 2019. *Accessible America*. New York: New York University Press.
- Williamson, Bess, and Elizabeth Guffey, eds. 2020. *Making Disability Modern: Design Histories*. London: Bloomsbury.
- Winner, Langdon. 1980. "Do Artifacts Have Politics?" *Daedalus* 109 (1): 121–136.
- Wong, Alice. 2015. "Assistive Technology by People with Disabilities, Part I: Introducing Team Free to Pee." *Model View Culture*, November 4, 2015. <https://modelviewculture.com/pieces/assistive-technology-by-people-with-disabilities-part-i-introducing-team-free-to-pee>.
- . 2019. "The Rise and Fall of the Plastic Straw: Sucking in Crip Defiance." *Catalyst: Feminism, Theory, Technoscience* 5 (1). <https://catalystjournal.org/index.php/catalyst/article/view/30435>.
- Yergeau, Melanie. 2014. "Disability Hacktivism." *Computers and Composition Online*. <http://cconlinejournal.org/hacking/>.